New evidence on the clinical, genetic, and biochemical bases of *GBA1*-Parkinson's disease: prospects for treatment

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Lancet Neurol 2026; 25: 602–14

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***GBA1* variants are common genetic risk factors for Parkinson's disease and also for dementia with Lewy bodies. New evidence highlights the relationship between the different *GBA1* variants and their associated clinical presentation and disease progression, although penetrance is low and the majority of carriers do not develop a synucleinopathy. The clinical profile of *GBA1*-associated Parkinson's disease is characterised by a faster rate of progression with more severe cognitive and autonomic dysfunction than idiopathic Parkinson's disease, particularly in those carrying severe pathogenic variants. The mechanisms involved in phenotypic conversion and disease progression in carriers of *GBA1* variants are being elucidated, and this knowledge could be translated into clinically relevant biomarker profiles. *GBA1* has become a target for intervention for both *GBA1*-associated Parkinson's disease and idiopathic Parkinson's disease, with therapeutic strategies aiming to prevent or slow disease progression via pharmacological chaperones, enzymatic allosteric activators, metabolic interventions, and modulation of gene expression.**

Introduction

Growing evidence highlights the limitations of conceptualising Parkinson's disease as a single, uniform clinical entity. Distinct clinical features, including prodromal presentation, biological signatures, and rate of progression, suggest the existence of multiple subtypes, each potentially driven by different underlying mechanisms. The inadequate characterisation of these subtypes constrains progress toward biological classification and targeted therapeutic strategies. Thus, a multidimensional framework that integrates genotype, α-synuclein seed amplification assays and other fluid biomarkers, neuroimaging, and longitudinal clinical measures could help shifting from broad descriptive subtypes to biologically grounded disease categories. An example of such an integrative framework is illustrated by evidence on the association between *GBA1* variants and Parkinson's disease. This association has been particularly transformative in elucidating pathogenic pathways, improving diagnostic precision, and enabling the development of targeted therapeutic approaches.

In this Review, we summarise current knowledge on the clinical, genetic, and biochemical features of *GBA1*-associated Parkinson's disease (hereafter referred to as *GBA1*-Parkinson's disease), with a focus on areas of ongoing debate and rapid progress, including findings from synuclein seed amplification assays and biomarkers of cognitive impairment. We then examine emerging hypotheses regarding the pathogenesis of *GBA1*-Parkinson's disease, such as contributions of mitochondrial dysfunction, alterations in glycosphingolipid metabolism, and the effect of peripheral *GBA1* expression on α-synuclein accumulation. Finally, we briefly review recent developments in clinical trials investigating *GBA1*-targeted therapeutic strategies.

Genetic and epidemiological findings

The *GBA1* gene encodes the lysosomal enzyme glucocerebrosidase, which catalyses the breakdown of glucosylceramide into ceramide and glucose in the lysosome.[1] The activation of glucocerebrosidase involves complex post-translational processing, including folding in the endoplasmic reticulum, glycosylation in the Golgi, trafficking to the endosomes and the lysosomes after binding the lysosomal integral membrane protein-2, and activation by saposin C, which derives from the cleavage of its precursor protein prosaposin.[1]

Biallelic pathogenic variants in *GBA1* result in deficient activity of glucocerebrosidase and consequent excessive accumulation of glucosylceramide, primarily in cells of the reticuloendothelial system, leading to the lysosomal storage disorder Gaucher disease, which can be classified into three types based on the severity of neurological symptoms (non-neuronopathic Gaucher disease type 1, and neuronopathic Gaucher disease types 2 and 3).[1] The most widely accepted classification of *GBA1* variants distinguishes four types of variants (figure 1[2–4]): mild, if reported in the homozygous state in Gaucher disease type 1; severe, if they cause Gaucher disease type 2 or 3; risk variants, if they do not cause Gaucher disease but are associated with increased risk for Parkinson's disease; and unknown, if there is insufficient information regarding Gaucher disease pathogenicity or Parkinson's disease risk.[4]

GBA1 variants are found in up to 10–15% of European patients with Parkinson's disease, with the prevalence increasing to 20% among patients of Ashkenazi Jewish descent.[5,6] Ethnic variation influences variant prevalence: the mild variant N409S is the most common variant in Jewish (Ashkenazi and non-Ashkenazi) populations; the severe variant L483P is the most prevalent in Asian and Hispanic populations; and the risk variant E365K is the most prevalent in individuals of European ancestry.[5,6] Recent studies have identified population-specific variants in approximately 50% of patients with Parkinson's disease who have African and African-admixed ancestry, such as the non-coding rs3115534-G variant. rs3115534-G is involved in splicing of *GBA1* mRNA and its presence leads to the reduction of protein expression and enzymatic activity levels.[7,8] The severe *GBA1* variants have the highest odds ratios (ORs) for

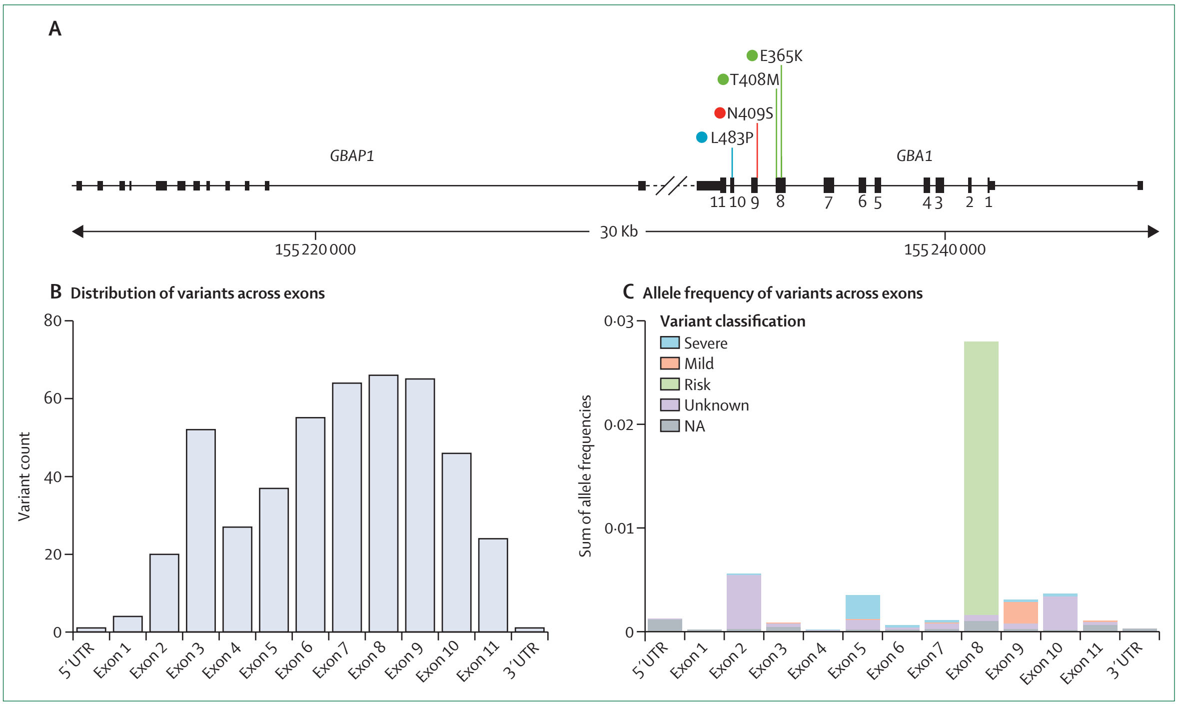

Figure 1: **Structure of the *GBA1* gene and distribution of *GBA1* variants**
(A) The *GBA1* gene is located on chromosome 1q22 and is composed of 11 exons and 10 introns, spanning a sequence of 7·6 Kb. Located 6·9 Kb downstream to the *GBA1* gene, there is the highly homologous untranslated pseudogene *GBAP1*. The homology is 96% in the coding region, and 98% between intron 8 and the 3′ UTR, where five identical segments longer than 200 bp each are present. The location of four relevant variants is highlighted and color-coded according to severity. Produced with Integrative Genomics Viewer.[2] (B) *GBA1* variants described in each exon according to Clinvar. Variants classified as benign and likely benign variants are excluded. (C) Distribution of *GBA1* variants across exons in the gnomAD database (healthy individuals),[3] and their classification into risk, mild, severe, variants of unknown significance, and NA.[4] NA=not available. UTR=untranslated region.

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Parkinson's disease (eg, L483P OR 6·4–30·4 and V433L OR 4·9–6·7), followed by the mild variants (eg, N409S OR 2·2–7·8), and the risk variants (eg, E365K OR 1·6–5·5).[4,5] Despite the associated increased risk, the penetrance of *GBA1* variants for Parkinson's disease is low and age-dependent, being estimated at 1·5–4·7% at age 60 years, and 7·7–9·1% at age 80 years.[5] Most heterozygous carriers of *GBA1* variants never develop Parkinson's disease over their lifetime. Similarly, the great majority of patients with Gaucher disease do not develop Parkinson's disease, even if they have a family history and despite carrying two pathogenic *GBA1* variants and having markedly lower levels of glucocerebrosidase activity.[9]

GBA1 variants are also associated with an increased risk of dementia with Lewy bodies, another α-synucleinopathy. Despite the clinical severity of dementia with Lewy bodies, mild and risk *GBA1* variants have also been identified in people with this condition,[10] highlighting the complexity of the link between *GBA1* variant severity and clinical presentation. The association between *GBA1* variants and isolated rapid eye movement (REM) sleep behaviour disorder—a condition associated with an 80–90% conversion rate to a synucleinopathy—has also been highlighted by large genome-wide association studies.[11]

Currently, there are no clinical tools to predict the likelihood of conversion to Parkinson's disease or dementia with Lewy bodies in *GBA1* variant carriers.[12] Several genetic and environmental modifiers influencing penetrance are under investigation. For example, recent studies indicate that a higher polygenic risk score independently contributes to increased risk for Parkinson's disease in *GBA1* variant carriers,[13,14] with carriers of pathogenic variants with a high score showing the highest cumulative incidence of Parkinson's disease.[13] From the clinical observation that carriers of both *GBA1* and *LRRK2* variants have a milder clinical phenotype compared with carriers of *GBA1* variants only, recent experimental work suggests that, in transgenic flies, expression of *LRRK2*^{mut} counteracts the pathological consequences (eg, neuroinflammation and loss of dopaminergic neurons) of overexpression of human *GBA1* variants.[15] Exposure to particular environmental toxicants (eg, pesticides) has been linked with increased risk of Parkinson's disease in *GBA1* variant carriers.[16,17] Further research is essential to refine the classification of *GBA1* variant carriers for their risk of Parkinson's disease, thus improving counselling strategies and enabling the accurate selection of carriers at high-risk for their inclusion in clinical trials of prevention interventions.

	Participants, N	Study design	Imaging technique	Main findings
Lee et al (2021)[23]	39 (*GBA1*-PD) *vs* 367 (iPD)	Longitudinal	[^{123}I]FP-CIT SPECT	Participants with *GBA1*-PD showed lower SBRs in the less-affected caudate nucleus
Chung et al (2021)[24]	54 (drug-naive *GBA1*-PD) *vs* 354 (drug-naive iPD)	Cross-sectional	[^{123}I]FP-CIT SPECT	No differences in striatal DAT between groups; higher MDS-UPDRS-III scores for the less affected side in patients with *GBA1*-PD, after adjusting for DAT availability in the contralateral putamen
Caminiti et al (2022)[25]	16 (GBA1-PD) *vs* 17 (early-onset iPD) *vs* 74 (late-onset iPD)*	Longitudinal	[^{123}I]FP-CIT SPECT	At baseline, *GBA1*-PD showed more marked [^{123}I]FP-CIT SUVr reduction in the striatal and extrastriatal regions than early-onset and late-onset iPD; at 2 years, no differences in global SUVr between groups
Grisanti et al (2023)[26]	25 (GBA1-PD) *vs* 25 (iPD)	Cross-sectional	[^{123}I]FP-CIT SPECT	*GBA1*-PD showed lower SBR in the most affected anterior putamen and left caudate
Kim et al (2023)[27]	16 (GBA1-PD) *vs* 24 (iPD)	Cross-sectional	[^{18}F]FP-CIT PET and [^{123}I]MIBG scintigraphy	*GBA1*-PD showed lower SUVr on bilateral posterior parietal cortex and part of the occipital cortex; no difference in striatal DAT; *GBA1*-PD showed delayed heart-to-mediastinum uptake

[^{18}F]FP-CIT PET= [^{18}F]-N-(3-fluoropropyl)-2β-carboxymethoxy-3β-(4-iodophenyl) nortropane PET. [^{123}I]FP-CIT SPECT=[^{123}I]ioflupane SPECT. [^{123}I]MIBG scintigraphy=[^{123}I] metaiodobenzylguanidine scintigraphy. *GBA1*-PD=*GBA1*-associated Parkison's disease. iPD=idiopathic Parkinson's disease. MDS-UPDRS-III=Movement Disorder Society Unified Parkinson's Disease Rating Scale III. SBR=specific binding ratio. SUVr=standard uptake value ratios. DAT=dopamine transporter. *Number of participants with available longitudinal data.

***Table 1:* Functional imaging studies comparing individuals with *GBA1*-Parkinson's disease with those with idiopathic Parkinson's disease**

GBA1-Parkinson's disease

The heterogeneity of clinical phenotypes and the absence of biomarkers correlating with clinical features are among the most challenging aspects of the study of *GBA1*-Parkinson's disease. In this section, we describe the clinical phenotype of *GBA1*-Parkinson's disease, emphasising its differences with idiopathic Parkinson's disease and how these differences might influence responses to therapy and, hence, can lead to management considerations in clinical practice. We then discuss how these clinical distinctions could be reflected by distinct biochemical profiles.

Clinical profile

The clinical profile of *GBA1*-Parkinson's disease is characterised by more pronounced motor and non-motor symptoms (particularly cognition, mood, sleep, and autonomic function) and a generally more aggressive disease course than idiopathic Parkinson's disease. These differences are most evident in carriers of severe variants, who have an earlier disease onset than individuals with idiopathic Parkinson's disease or carriers of other variants.[6,18–21]

Carriers of severe variants have more frequent motor complications, such as motor fluctuations and dyskinesias.[6] It has been shown that DAT values are substantially lower in patients with Parkinson's disease with dyskinesias.[22] However, this pattern is not consistently reflected in *GBA1*-Parkinson's disease dopaminergic imaging (table 1).[23–27] Cognitive impairment is more frequent and severe in people with *GBA1*-Parkinson's disease than in those with idiopathic Parkinson's disease,[19,28] especially in carriers of severe variants.[21] A longitudinal study has revealed earlier cognitive deterioration in *GBA1*-Parkinson's disease than idiopathic Parkinson's disease,[25] with more pronounced in cortical thinning of posterior brain regions and frontal and orbitofrontal cortices, but with similar patterns of subcortical, hippocampal, and amygdala volume loss.[29] Mood disorders are more frequent in *GBA1*-Parkinson's disease than in idiopathic Parkinson's disease, especially in carriers of severe variants.[21] A 2022 meta-analysis identified an elevated risk of REM sleep behaviour disorder in patients with *GBA1*-Parkinson's disease, notably among carriers of the severe L483P or the mild N409S variants.[30] An increased prevalence of orthostatic hypotension has been reported in many cohort studies of people with *GBA1*-Parkinson's disease, compared with patients with idiopathic Parkinson's disease,[20,27,31] but no differences were reported in one case–control study[28] and in another study in which orthostatic hypotension was assessed by cardiovascular reflex tests.[31] A few studies reported impairment in the parasympathetic regulation of cardiovascular function,[28,32] in association with the presence of REM sleep behaviour disorder and constipation.[28] These findings align with imaging data showing more prominent cardiac sympathetic denervation in *GBA1*-Parkinson's disease than idiopathic Parkinson's disease.[27] Overall, the *GBA1*-Parkinson's disease phenotype (particularly in carriers of severe variants) is marked by more pronounced autonomic dysfunction, REM sleep behaviour disorder, and faster cognitive decline, and seemingly aligns more closely with the body-first Parkinson's disease subtype than with the brain-first subtype.[27]

Given their frequent motor complications and also the complex non-motor profile that patients with *GBA1*-Parkinson's disease have, growing attention has been directed toward evaluating device-aided therapies in these patients. An early retrospective study pooling data from

	Study	Participants, N	Study design	Biospecimen (method)	Main findings
Protein					
α-synuclein	Lerche et al (2025)[51]	129 (*GBA1*-PD) *vs* 188 (iPD)	Cross-sectional and longitudinal	CSF (SAA RT-QuIC)	91% positive in *GBA1*-PD; 93% positive in iPD
α-synuclein	Brockmann et al (2021)[52]	99 (*GBA1*-PD; 53 risk, 17 mild, 29 severe) *vs* 107 (iPD)	Cross-sectional and longitudinal	CSF (SAA RT-QuIC)	87% positive in *GBA1*-PD (91% in risk, 65% in mild, and 93% in severe); 91% positive in iPD; shortest TTT and highest AUC in severe *GBA1*-PD compared with other genetic variants
α-synuclein	Orrú et al (2025)[53]	PPMI cohort: 76 (*GBA1*-PD) *vs* 764 (iPD); Tübingen cohort: 83 (*GBA1*-PD) *vs* 98 iPD	Cross-sectional and longitudinal	CSF (PPMI cohort: SAA PMCA); CSF (Tübingen cohort: SAA RT-QuIC)	PPMI: 92% positive in *GBA1*-PD positive; 92% positive in iPD; faster TTT in *GBA1*-PD; Tübingen: faster TTT and higher AUC in *GBA1*-PD
α-synuclein	Huh et al (2021)[43]	44 (*GBA1*-PD) *vs* 227 (iPD)	Cross-sectional (with available longitudinal clinical data)	CSF (ELISA)	No differences
α-synuclein	Lerche et al (2021)[45]	Tübingen cohort: 102 (*GBA1*-PD) *vs* 414 (iPD); PPMI cohort: 34 (*GBA1*-PD) *vs* 373 (iPD)	Cross-sectional and longitudinal	CSF (ELISA)	Lower concentrations in *GBA1*-PD
α-synuclein	Avenali et al (2021)[42]	29 (*GBA1*-PD) *vs* 37 (iPD)	Cross-sectional	PBMCs (ELISA)	Higher concentrations in *GBA1*-PD (higher in severe *vs* risk variant)
Lysosomal markers*					
Glucocerebrosidase activity	Lerche et al (2021)[45]	Tübingen cohort: 89 (*GBA1*-PD) *vs* 41 (iPD); PPMI cohort: 31 (*GBA1*-PD) *vs* 277 (iPD)	Cross-sectional and longitudinal	Tübingen: fresh blood leukocytes; PPM1: DBS	More reduced in *GBA1*-PD (lower in severe compared with risk variants)
Glucocerebrosidase activity	Avenali et al (2021)[42]	29 (*GBA1*-PD) *vs* 37 (iPD)	Cross-sectional	PBMCs	More reduced in *GBA1*-PD (lower in mild and severe compared with risk variants)
Glucocerebrosidase activity	Omer et al (2022)[44]	102 (*GBA1*-PD) *vs* 30 (iPD)	Cross-sectional	DBS	More reduced in *GBA1*-PD
Glucocerebrosidase activity	Kopytova et al (2022)[41]	18 (*GBA1*-PD) *vs* 177 (iPD)	Cross-sectional	DBS	More reduced in *GBA1*-PD
Glucocerebrosidase activity	Thaler et al (2021)[47]	77 (*GBA1*-PD) *vs* 31 (iPD)	Cross-sectional	DBS	More reduced in *GBA1*-PD
Glucocerebrosidase-related lysosomal proteins	Avenali et al (2021)[42]	29 (*GBA1*-PD) *vs* 37 (iPD)	Cross-sectional	PBMCs	Higher LIMP-2 and lower saposin C concentrations in *GBA1*-PD
BMP	Merchant et al (2023)[38]	76 (*GBA1*-PD; N409S) *vs* 379 (iPD)	Cross-sectional and longitudinal	Urine	*GBA1*-PD showed higher concentrations of total di-22:6 and 2,2′-di-22:6 BMP at baseline; no difference in total di-18:1 BMP; BMP isoforms did not predict decline in striatal dopaminergic imaging or motor or cognitive function over 5 years

(Table 2 continues on next page)

12 datasets reported that *GBA1* variant carriers who underwent deep brain stimulation, mainly targeting the subthalamic nucleus, not only had faster cognitive decline up to 5 years following neurosurgery than those with idiopathic Parkinson's disease who underwent neurosurgery, but also than those with *GBA1*-Parkinson's disease who were not treated with deep brain stimulation.[33] Within the *GBA1*-Parkinson's disease group who underwent deep brain stimulation, those individuals with mild or severe variants declined faster than those with risk variants.[33] Another retrospective, multicentre study analysing 5-year follow-up data reported that cognitive function declined more rapidly in patients with *GBA1*-Parkinson's disease (8 [25%] of 32) than in those with idiopathic Parkinson's disease (15 [11%] of 140).[20] In another study, individuals with *GBA1*-Parkinson's disease had a significantly increased risk for cognitive decline and psychotic events 10 years after deep brain stimulation.[34] All of the above-mentioned studies report similar substantial improvements in motor function and reduction of motor complications after deep brain stimulation in both patients with idiopathic Parkinson's disease or *GBA1*-Parkinson's disease.[20,33,34] A recent study has contradicted these results, showing that patients with *GBA1*-Parkinson's disease have a higher risk of developing dementia regardless of treatment with deep brain stimulation, and compared with patients with idiopathic Parkinson's disease who received deep brain stimulation.[35] No relevant differences were detected when patients were stratified according to *GBA1* variant severity.[35] Regarding levodopa–carbidopa intestinal gel infusion, two recent multicentre retrospective studies showed that patients with *GBA1*-Parkinson's disease required infusion earlier in their disease course and had more frequent hallucinations, faster cognitive decline, worse orthostatic hypotension, and less efficacy on motor complications

	Study	Participants, N	Study design	Biospecimen (method)	Main findings
(Continued from previous page)					
Glycosphingolipids and ceramides*					
Glucosylceramide, glucosylsphingosine, and galactosylsphingosine	den Heijer et al (2023)[39]	48 (*GBA1*-PD) *vs* 8 (iPD)	Cross-sectional	Plasma and PBMCs	*GBA1*-PD had higher plasma glucosylceramide (multiple isoforms) and glucosylsphingosine concentrations; no differences in plasma galactosylsphingosine; in PMBCs, no differences in glucosylceramide, glucosylsphingosine, and galactosylsphingosine concentrations
Glucosylsphingosine d18:1	Surface et al (2022)[40]	20 (*GBA1*-PD; N409S) *vs* 20 (iPD)	Cross-sectional	Plasma	Higher concentrations in *GBA1*-PD
HexSph (including glucosylsphingosine and galactosylsphingosine)	Kopytova et al (2022)[41]	18 (*GBA1*-PD) *vs* 177 (iPD)	Cross-sectional	DBS	Higher concentrations in *GBA1*-PD
Glucosylceramide, ceramides, and sphingomyelins	Lerche et al (2021)[45]	Tübingen cohort: 51 (*GBA1*-PD) *vs* 45 (iPD); PPMI cohort: 31 (*GBA1*-PD) *vs* 276 (iPD)	Cross-sectional and longitudinal	CSF	Higher levels of total amount of glucosylceramides and several sphingomyelin species in *GBA1*-PD; higher levels of total amount of ceramides in severe variants
Glucosylceramide, sphingomyelin, ceramides	Huh et al (2021)[43]	44 (*GBA1*-PD) *vs* 227 (iPD)	Longitudinal	CSF	Higher glucosylceramide and reduced sphingomyelin in *GBA1*-PD; no difference in ceramides
d18:1 sphingolipid species	Lerche et al (2024)[50]	84 (*GBA1*-PD + *GBA1*-DLB) *vs* 105 (iPD + iDLB)	Cross-sectional and longitudinal	CSF	No differences
Inflammatory markers					
Cytokines and chemokines	Galper et al (2021)[46]	135 (*GBA1*-PD) *vs* 75 (iPD)	Cross-sectional	Plasma	No differences
Cytokines	Thaler et al (2021)[47]	77 (*GBA1*-PD) *vs* 31 (iPD)	Cross-sectional	Serum and CSF	No differences
Oxidative stress markers					
Uric acid	Koros et al (2021)[48]	120 (*GBA1*-PD) *vs* 369 (iPD)	Longitudinal	Serum	No differences at baseline and longitudinally
Uric acid	Thaler et al (2021)[49]	87 (*GBA1*-PD) *vs* 105 (iPD)	Cross-sectional	Serum	No differences

AUC=area under the fluorescence curve during the reaction time. BMP=Bis(monoacylglycerol)phosphate. DLB=dementia with Lewy bodies. DBS=dried blood spots. *GBA1*-PD=*GBA1*-associated Parkinson's disease. GD=Gaucher disease. HexSph=hexosylsphingosine. iPD=idiopathic Parkinson's disease. LIMP-2=lysosomal integral membrane protein 2. PBMC=peripheral blood mononuclear cells. PD=Parkinson's disease. PMCA=protein misfolding cyclic amplification. PPMI=Parkinson's Progression Markers Initiative. RT-QuIC=real-time quaking-induced conversion. SAA=synuclein seeding amplification assay. TTT=time to threshold. *Studies on glucocerebrosidase activity and glycosphingolipids did not show a difference between *GBA1*-Parkinson's disease and *GBA1* variant carriers without Parkinson's disease.

***Table 2:* Biomarkers studies comparing *GBA1*-Parkinson's disease with idiopathic Parkinson's disease**

than patients with idiopathic Parkinson's disease who received the same treatment.[36,37]

Based on this contrasting evidence, genetic testing for *GBA1* has not yet been included in the prescreening evaluation for device-aided therapies. Nonetheless, in individuals with early disease onset and marked non-motor impairment, *GBA1* genetic screening might be advisable. No clear recommendations can be made at present as prospective, controlled studies are needed. However, if genetic screening returns a positive result, information about natural progression of *GBA1*-Parkinson's disease should be shared with patients and their families alongside details about the risks and benefits of each device-aided therapy, to manage prognosis and potential complications.

Biochemical profile

The prospect of identifying a biochemical signature for the severe phenotype associated with *GBA1*-Parkinson's disease, particularly with fast cognitive decline, has garnered considerable interest.[38–53] The most exciting results come from studies exploring synuclein seed amplification assays (SAAs) in *GBA1*-Parkinson's disease. Among the different seeding kinetic measures, faster time to threshold (TTT) and higher area under the fluorescence curve (AUC) during the reaction time were detected in large cohort studies of patients with *GBA1*-Parkinson's disease compared with patients with idiopathic Parkinson's disease (table 2).[53] Notably, accelerated seeding kinetics predicted cognitive decline independently of Alzheimer's disease co-pathology, suggesting that heightened seeding capacity could reflect a faster cortical spread of Lewy body pathology in *GBA1*-Parkinson's disease.[53] These results temper the potential role played by Alzheimer's disease co-pathology in cognitive decline affecting *GBA1*-Parkinson's disease. Indeed, similar CSF levels of biomarkers of Alzheimer's disease pathology (Aβ42 and tau phosphorylated at position 181) were identified in *GBA1*-Parkinson's disease and idiopathic Parkinson's disease, with no *GBA1*-Parkinson's disease cases showing a combined CSF α-synuclein and Alzheimer's disease profile, and only 2% (n=2) of cases (who were carriers of the mild N409S variant) showing an isolated Alzheimer's disease profile.[51] The reason for this finding remains unknown. Similarly, large post-mortem studies have showed that *GBA1* variants were more frequent in cases with higher burden of Lewy body pathology and lower burden of Alzheimer's disease pathology.[54] Although no analyses stratified by variant types were done in a recent SAA kinetics study,[53] a previous study using real-time quaking-induced conversion (RT-QuIC) analysis of CSF revealed that carriers of severe variants had the shortest TTT and highest AUC based on RT-QuIC kinetic parameters, compared with carriers of different genetic variants.[52] Further evidence suggesting that Alzheimer's disease co-pathology does not contribute to cognitive decline in *GBA1*-Parkinson's disease includes findings that the *APOE* ε4 allele is not associated with cognitive deterioration or Alzheimer's disease CSF biomarkers in *GBA1*-Parkinson's disease, including for patients carrying severe variants who develop cognitive impairment early in their disease course.[55] Moreover, a large post-mortem study on patients with dementia with Lewy bodies found no association between *APOE* ε4 status and isolated Lewy bodies pathology.[56]

Endolysosomal markers such as glucocerebrosidase activity have shown little utility as markers of disease severity. Glucocerebrosidase activity is consistently lower in patients with *GBA1*-Parkinson's disease than idiopathic Parkinson's disease in dried blood spots and peripheral blood mononuclear cells,[41,42,44,45,47] with increased variant severity associated with lower levels of glucocerebrosidase.[42] Very low enzymatic activity is suggestive of Gaucher disease, but very few of these individuals develop Parkinson's disease.

Measurements of concentrations of enzymatic substrates (glucosylceramide and glucosylsphingosine) and the downstream products of the broader sphingolipid pathway have also poor utility as biomarkers of disease severity. Slightly higher plasma concentrations of glucosylsphingosine have been reported in *GBA1*-Parkinson's disease compared with idiopathic Parkinson's disease.[40,41] However, glucosylsphingosine concentrations detected in *GBA1*-Parkinson's disease were similar to those found in carriers without Parkinson's disease,[40] and fell within the normal range for healthy individuals;[40,41] thus, glucosylsphingosine is not a clinically relevant or biologically meaningful biomarker for diagnosis or progression monitoring.[57,58] Glucosylceramide was found to be elevated in *GBA1*-Parkinson's disease compared with idiopathic Parkinson's disease, in both plasma and CSF,[39,43] but no significant associations between longitudinal concentrations and cognitive or motor function were detected in early-stage *GBA1*-Parkinson's disease.[43] Cognitive, but not motor, function declined more rapidly in people with *GBA1*-Parkinson's disease in the highest quartile of CSF glucosylceramide to sphingomyelin ratio, compared with those in the lowest quartile, suggesting a possible role of glucosylceramide in cognitive decline. Yet, data from a clinical trial showing no changes in cognitive function despite reduction of CSF concentrations of glucosylceramide undermine this hypothesis.[59] Overall, current evidence supports the hypothesis that the severe clinical phenotype and accelerated cognitive decline observed in *GBA1*-Parkinson's disease arise from a predominant α-synuclein pathology, driven by increased α-synuclein seeding activity.

The role of glucocerebrosidase deficiency in pathogenesis

Building on the distinct clinical features of *GBA1*-Parkinson's disease, increasing attention has turned to the underlying molecular mechanisms, particularly the role of glucocerebrosidase deficiency in driving pathogenesis, including the accumulation and spread of α-synuclein. Several steps along the secretory pathway of

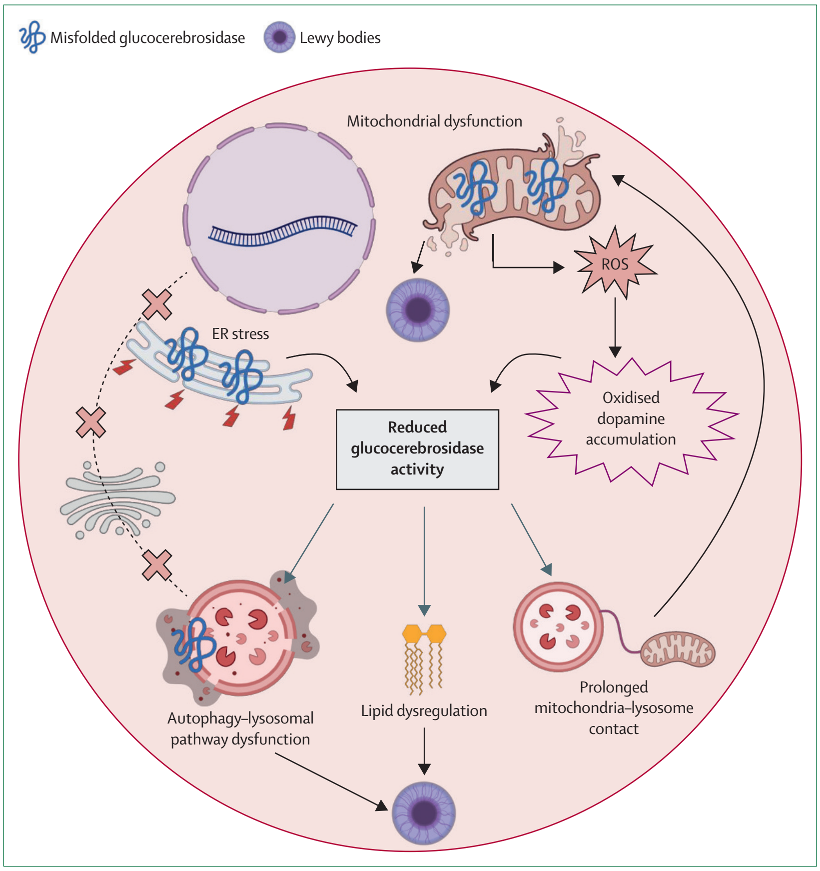

Figure 2: **Cellular pathways implicated in *GBA1*-Parkinson's disease**
Genetic variants in the *GBA1* gene lead to alterations in the multistep post-translational processing of the glucocerebrosidase protein through the ER and Golgi apparatus and its final transportation to the lysosome (pathway marked by a black dotted line interrupted by red crosses). Different genetic variants have been associated with different dysfunctional mechanisms, although pathways might act synergistically. Misfolded glucocerebrosidase can accumulate in the ER and the mitochondria, leading to increased stress in both ER for severe variants and mitochondria for risk and severe variants. As a result, the enzymatic activity of glucocerebrosidase is reduced. This reduction has several consequences at the cellular level, including alterations in the autophagy system, changes in lipid membrane composition, and prolongation of mitochondria-lysosome contacts. Prolonged mitochondria-lysosome contacts can further impair mitochondrial function for mild and severe variants. Impaired mitochondrial metabolism can reduce glucocerebrosidase activity by increasing intracellular oxidised dopamine. Misfolded glucocerebrosidase can also bind the lysosomal membranes, thus impairing the autophagic machinery for mild and severe variants. This multiorganelle dysfunction results in alterations of the soluble conformation of α-synuclein, with formation of insoluble, fibrillar components, and Lewy bodies. ER=endoplasmic reticulum. ROS=reactive oxygen species.

glucocerebrosidase can be disrupted by *GBA1* variants resulting in loss-of-function (loss of or reduced enzymatic activity) or gain-of-function, including effects on stress responses, with variant-specific mechanisms. For instance, the frameshift mutation arising from 84_85insGG, which causes a shift at L29fs, does not result in the translation of a mutant protein, but instead leads to downstream effects primarily due to loss of function. Alternatively, the severe L483P variant, and to a lesser extent the mild N409S variant, but not the risk E365K variant, cause misfolding of glucocerebrosidase in the endoplasmic reticulum, leading to substantial retention of the misfolded protein, associated degradation, and stress in the endoplasmic reticulum.[60,61] These different mechanisms, which might contribute synergistically to *GBA1*-Parkinson's disease, are described in the following sections and summarised in figure 2.

Panel: Emerging *GBA1*-targets for therapeutic approaches

mTOR complex 1[67,69]
- Negative regulator of the autophagy–lysosomal pathway
- Hyperactivation of the mTOR pathway has been detected in GBA1mut experimental models
- Inhibitors of mTOR were able to inhibit phosphorylation of α-synuclein in *GBA1*-Parkinson's disease dopaminergic neurons
- Stimulation of autophagy with mTOR inhibitor rapamycin partly rescued the glucocerebrosidase-deficient phenotype in *GBA1* knock-out flies

Prosaposin C[70]
- Precursor of saposin C, a lysosomal protein acting as activator of glucocerebrosidase
- Overexpression of prosaposin C reduced monomeric α-synuclein concentrations in SH-SY5Y cells
- Recombinant saposin C detached α-synuclein from artificial glucocerebrosidase-enriched lipid membranes at lysosomal pH

Progranulin[71]
- Protein acting as a modifier of glucocerebrosidase activity by binding with HSP70
- Progranulin deficiency promoted decreased glucocerebrosidase activity in lysosomes, lipid accumulation, and increased accumulation of phosphorylated α-synuclein
- A brain-penetrant progranulin-derived peptide prevented α-synuclein accumulation in double-mutant *GRN*^{mut} and *Gba1*^{mut} mice

Cathepsin B[72,73]
- Cysteine protease involved in lysosomal protein degradation, autophagy, and lysosome biogenesis regulation, and cleavage of monomeric and fibrillar α-synuclein
- Cathepsin B inhibition impaired glucocerebrosidase activity in lysosomes in dopaminergic neurons
- Increased expression of CTSB gene enhanced α-synuclein fibril clearance

Cathepsin D[74]
- Aspartyl protease involved in proteolytic degradation, cell invasion, and apoptosis
- Recombinant human proform of cathepsin D reduced insoluble concentrations of α-synuclein in experimental models
- In neuroglioma cells overexpressing α-synuclein, treatment with recombinant cathepsin D rescued glucocerebrosidase activity in lysosomes

Cathepsin L[75]
- Cysteine protease implicated in glucocerebrosidase stability
- Chemical inhibition of cathepsin L increased glucocerebrosidase concentrations and reduced phosphorylated 129-α-synuclein in dopaminergic neurons from people with *GBA1*-Parkinson's disease

TMEM175[76]
- Proton and K+ selective channel located on lysosomal membranes, implicated in maintaining lysosomal homoeostasis
- Loss of TMEM175 function diminishes channel conductance, causing excessive lysosomal acidification, reduced proteolytic efficiency, disrupted autophagic flux, and ultimately promoting α synuclein aggregation
- TMEM175 agonists facilitate the degradation of pathological α-synuclein by promoting lysosomal removal in neurons

The autophagic machinery

Glucocerebrosidase deficiency can disrupt the autophagy–lysosomal pathway, promoting α-synuclein accumulation.[62] Studies suggest that the underlying mechanisms vary, with autophagic responses differing according to *GBA1* variant type and model system.[62] For instance, a study using dopaminergic neurons derived from induced pluripotent stem cells showed that mild variants affected the chaperone-mediated autophagy (CMA) pathway, thus increasing the monomeric fraction of α-synuclein, whereas severe variants affected macroautophagy and the homoeostasis of the endoplasmic reticulum, thus resulting in accumulation and release of oligomeric α-synuclein.[63]

CMA is a selective form of autophagy in which proteins (including α-synuclein) are recognised by the cytosolic chaperone HSC70 and then transferred into the lysosome, where they are recognised by the lysosomal receptor LAMP-2A.[64] In post-mortem human brains from people with *GBA1*-Parkinson's disease, a fraction of misfolded mutant GBA was shown to be bound to the lysosomal membrane; this mislocalisation depended on a pentapeptide motif in mutant GBA recognised by HSC70.[64] Mutant GBA at the lysosomal surface disrupted CMA, causing accumulation of CMA substrates, including α-synuclein.[64] In line with this finding, lysosomal concentrations of LAMP-2A were decreased in *GBA1*^{N409S} and *GBA1*^{L483P} differentiated human dopaminergic-like neuroblastoma cells compared with differentiated human dopaminergic-like neuroblastoma cells.[61]

Alternative mechanisms mediating dysfunction of the autophagy–lysosomal pathway in *GBA1*-Parkinson's disease might involve the mTOR complex 1, which negatively regulates this pathway through the control of the nuclear translocation of transcription factor EB.[65] Hyperactivation of mTOR complex 1 has been associated with glucocerebrosidase deficiency, possibly as a consequence of substrate accumulation, in several studies using dopaminergic neurons from patients with *GBA1*-Parkinson's disease carrying severe or mild variants.[66–68] In these models, inhibition of mTOR could reverse α-synuclein aggregation and reduce endoplasmic reticulum stress,[66,67] making the mTOR pathway a potential therapeutic target in *GBA1*-Parkinson's disease, although the exact role of mTOR complex 1 activation and its contribution to *GBA1*-Parkinson's disease is still not fully understood (panel). Finally, glucocerebrosidase inhibition has been also shown to directly impair the protein and sphingolipid composition of the plasma membrane,[77] thus potentially affecting the maturation and fusion of autophagosomes and endosomes.[78]

Mitochondrial dysfunction

Mitochondrial dysregulation and oxidative stress have also been implicated in *GBA1*-Parkinson's disease, with emerging evidence linking these processes to enhanced α-synuclein accumulation and propagation.[79,80]

In the mitochondria, glucocerebrosidase can promote the integrity and function of mitochondrial complex I.[80] Using dopaminergic neurons and differentiated midbrain organoids derived from induced pluripotent stem cells from patients with *GBA1* mutations, investigators detected alterations in respiratory chain complex I integrity and cellular energy metabolism.[80] Interestingly, *GBA1* variants in mouse brains caused enhanced generation of reactive oxygen species and accumulation of reactive nitrogen species, which promoted α-synuclein aggregation.[79] Impaired mitophagy (ie, the selective removal of dysfunctional mitochondria by autophagy) has also been shown in several $GBA1^{mut}$ models.[81]

Dysfunctional mitochondria and progressive accumulation of oxidised dopamine were shown to contribute to decreased wild-type glucocerebrosidase activity and lysosomal dysfunction in dopaminergic neurons derived from people with idiopathic Parkison's disease or with genetic forms of the disease, suggesting a novel mechanism that at least partly explains the preferential vulnerability of dopaminergic neurons across different forms of Parkinson's disease.[62]

Finally, the potential role of mitochondria–lysosome contacts as an upstream regulator of mitochondrial function and dynamics in *GBA1*-Parkinson's disease has also been proposed. Mitochondria–lysosome contact formation is a process promoted by the activity of GTPase-bound Rab7, located on the lysosomal membrane, while contact untethering is mediated by recruitment of the RAB7GTPase-activating protein TBC1D15 to mitochondria; in *GBA1*-Parkinson's disease-derived dopaminergic neurons, prolonged mitochondria–lysosome contact tethering was due to dysregulation of TBC1D15, resulting in defects in mitochondrial distribution and function.[82] Interestingly, these defects could be rescued by the expression of human TBC1D15 in $GBA1^{mut}$ neurons.[82]

Lipid metabolism

Alterations in sphingolipid and metabolism of other lipids have also been implicated in *GBA1*-Parkinson's disease, offering further insight into the complex biochemical landscape underlying its pathogenesis. The enzymatic substrates (glucosylceramide and glucosylsphingosine) can mediate the transformation of α-synuclein into toxic forms.[67] The absence of evidence for substrate accumulation in post-mortem brains of individuals with *GBA1*-Parkinson's disease has placed the pathogenic role of this substrate accumulation under question,[1] but local accumulation within the lysosomal membrane could still contribute to pathogenesis.

Ceramides are products of the glucocerebrosidase pathway and important components of lipid membrane homoeostasis. Inhibition of acid ceramidase, the lysosomal enzyme responsible for the degradation of ceramide into sphingosine and free fatty acids, decreased α-synuclein concentrations in dopaminergic neurons derived from patients with *GBA1*-Parkinson's disease, possibly by correcting lysosomal lipid alteration and by preventing mTOR hyperactivity and restoring autophagic flux.[67] Ceramides can activate cathepsin B, which promotes the cleavage of prosaposin to saposin C in lysosomes,[72] and can cleave α-synuclein in lysosomes,[73] so it is possible that ceramide concentrations can influence α-synuclein by regulation of cathepsin B levels. However, more recent work on post-mortem CSF and tissues from patients with dementia with Lewy bodies (with and without *GBA1* variants) showed increased ceramide concentrations, α-synuclein, and tau proteins in extracellular vesicles purified from CSF and frontal cortex, regardless of *GBA1* genotype,[83] so the association between ceramide and α-synuclein levels remains unclear. Interestingly, the enzymes β galactosylceramidase and acid sphingomyelinase, respectively encoded by *GALC* and *SMPD1*, are involved in ceramide metabolism and genetic variants in these genes have been associated with Parkinson's disease.[84]

Changes in lipid membrane composition can alter α-synuclein aggregation propensity, and this process can be modulated by glucocerebrosidase.[85] Studies conducted on fibroblasts from patients with $GBA1^{L483P}$ Parkinson's disease showed a substantial increase in the proportion of sphingomyelin, ceramide, and short chain (C34) hexosylceramide molecules and a decrease of molecules with long chain (C42). The increased C34:C42 sphingolipid ratio correlated with reduced glucocerebrosidase activity. Lipid extracts from patients with *GBA1*-Parkinon's disease triggered in vitro fibrillation of recombinant human α-synuclein more effectively than lipids from controls with no history of neurological disorders.[85] Within the cell, lipid metabolism is regulated through the turnover and recycling of lipids in membranes and organelles such as lipid droplets.[78] Studies using RNA sequencing have revealed that the pharmacological inhibition of glucocerebrosidase can upregulate the expression of genes involved in coating lipid droplets.[86] Increased lipid droplet formation was observed in SH-SY5Y cells overexpressing the E365K variant and in E365K-mutant fibroblasts, albeit without changes in glucocerebrosidase enzymatic activity.[60] This finding suggests that, at least for some *GBA1* variants, the link between lipid metabolism and

GBA1-Parkinson's disease might extend beyond enzymatic activity.

α-synuclein spreading

Interneuronal spread of misfolded α-synuclein is considered an important pathogenic step that might underlie the progressive exacerbation of α-synuclein pathology during Parkinson's disease development and progression. It has been shown that glucocerebrosidase deficient cells treated with preformed fibrils increase the release of α-synuclein in mouse cortical neurons, and similarly, injection of mouse α-synuclein preformed fibrils into the striatum of mice that carry the L483P knock-in variant caused significantly greater spread of α-synuclein than injection in wild-type littermates.[78] However, following injection of α-synuclein preformed fibrils in the olfactory bulb of young mice harbouring the D448V variant, the spread of α-synuclein pathology was not accelerated, even 6 months after injection.[87] It is noteworthy, however, that D448V is not a Parkinson's disease-linked mutation and, for this reason, data derived from this animal model warrant cautious interpretation. In a different in vivo model of Parkinson's disease in which α-synuclein is overexpressed in the medullary vagal system, caudo-rostral α-synuclein brain spreading was exacerbated in L483P-carrying transgenic mice, likely due to mitochondrial oxidant stress.[79,88] In summary, several cell and animal models support a role for glucocerebrosidase deficiency in promoting α-synuclein spread. However, there is no direct evidence of this relationship in human beings.

The clinical phenotypes of some patients with *GBA1*-Parkinson's disease align with the body-first hypothesis of Parkinson's disease, according to which α-synuclein pathology originates in the enteric nervous system and then propagates caudorostrally to the autonomic nervous system and the CNS.[89] How *GBA1* dysfunction could contribute to gastrointestinal α-synuclein accumulation and potential gut-to-brain spread still needs to be clarified. Flies lacking the *Drosophila* orthologue of *GBA1* show altered gut permeability, microbiome composition, and gastrointestinal function.[69] Gut microbiome components were found to increase α-synuclein concentrations in mouse gut enteroendocrine cells.[90] Considering that ingested toxicants or inflammatory stimuli can cause gastrointestinal α-synuclein pathology, a recent hypothesis proposes that *GBA1* dysfunction might act alongside these factors to drive *GBA1*-Parkinson's disease.[91]

GBA1-targeted clinical trials

Understanding the molecular mechanisms underlying *GBA1*-Parkinson's disease has paved the way for the development of therapeutic strategies aimed at restoring glucocerebrosidase function, hence mitigating downstream pathogenic processes. We describe the clinical evidence supporting therapeutic approaches targeting *GBA1*, focusing on those whose investigation has progressed into clinical trials.

Substrate reduction therapy

Despite the conflicting evidence of substrate accumulation in *GBA1*-Parkinson's disease, reduction of substrate accumulation by direct inhibition of glucosylceramide synthase has been tested as a disease-modification strategy. Based on murine studies, and after showing a good safety and tolerability profile, target engagement, and brain penetration in a phase 1 trial,[92,93] the glucosylceramide synthase inhibitor venglustat has been investigated in a phase 2, multicentre, randomised controlled trial (RCT; MOVES-PD) in participants with *GBA1*-Parkinson's disease (table 3). Venglustat showed good target engagement, with about 75% reduction in CSF and plasma concentrations of glucosylceramide from baseline.[59] No beneficial clinical effect of venglustat was shown compared with placebo.[59] It remains unclear whether the ineffectiveness of venglustat, established early during the trial, might have resulted from an anti-dopaminergic effect (eg, mediated through altered dopaminergic drug metabolism or impaired dopamine release), or whether this finding challenges the hypothesis of a therapeutic role for substrate reduction in the clinical progression of *GBA1*-Parkinson's disease.[59] Other brain-penetrant glucosylceramide synthase inhibitors, such as GZ667161, have not reached clinical trials yet.

Molecular chaperones and allosteric activators

Pharmacological chaperones of glucocerebrosidase are small molecules that can increase enzymatic function via promotion of correct folding and trafficking of the protein to the lysosome, thus enhancing lysosomal concentrations of glucocerebrosidase and reducing misfolded protein, whose accumulation impairs several cellular pathways, including endoplasmic reticulum stress, mitochondrial disfunction, and decreased CMA.[1,96]

The most studied pharmacological chaperone in *GBA1*-Parkinson's disease is ambroxol. Functioning as a mixed-type chaperone (with maximal inhibitory activity in the neutral pH of the cytoplasm, intermediate in the endoplasmic reticulum, and undetectable in the acidic pH of the lysosome), ambroxol showed positive effects in several in vitro and in vivo models.[1] The initial testing of ambroxol in a phase 2, open-label trial (AiM-PD) was successful, with ambroxol increasing CSF concentrations of glucocerebrosidase and total α-synuclein, and improving motor function at 6 months, although caution in the interpretation of the positive clinical outcome is required due to the design of the study and the small sample size.[1] No effect on cognition, despite target engagement, was detected in another phase 2 RCT testing ambroxol in individuals with mild to moderate Parkinson's disease and dementia.[94] Ambroxol is also under evaluation in two phase 2 RCTs, one in patients with *GBA1*-Parkinson's disease (NCT05287503) and the other in

	Pathway	Trial	Phase	Participants (N)	Recruitment status	Primary outcome	Results*
Venglustat (15 mg/day) *vs* placebo	Substrate reduction	NCT02906020 (MOVES-PD)	2	221 *GBA1*-PD (110 venglustat *vs* 111 placebo)	Completed: terminated early	Safety, tolerability, and changes in MDS-UPDRS part II+III score at week 52 *vs* baseline	Good safety and tolerability; no benefit in motor function[59]
Ambroxol (1260 mg/day)	Chaperone	NCT02941822 (AiM-PD)	2†	17 PD (of which 8 *GBA1*-PD)	Completed	Changes in CSF ambroxol concentration and Glucocerebrosidase activity at baseline *vs* 186 days	↑ ambroxol concentration; ↑ glucocerebrosidase protein; ↑ total α-synuclein; ↓ MDS-UPDRS part III[1]
Ambroxol (2 dose groups: 1050 mg/day, 525 mg/day) *vs* placebo	Chaperone	NCT02914366	2	54 PDD (22 on 1050 mg/day ambroxol *vs* 24 placebo; 8 on 525 mg/day excluded from final analyses)	Completed	Changes in ADAS-Cog-13 and CGIC scores	No changes in clinical outcomes; ↑ glucocerebrosidase activity in lymphocytes in ambroxol group[94]
Ambroxol (1200 mg/day) *vs* placebo	Chaperone	NCT05287503 (AMBITIOUS)	2	65 *GBA1*-PD	Completed; awaiting results	Changes in MOCA score and cognitive status at baseline *vs* week 52	NA
Ambroxol (1260 mg/day) *vs* placebo	Chaperone	NCT05778617 (ASPro-PD)	3a	330 PD with or without *GBA1* variants (target)	Active	Changes in MDS-UPDRS part I–III score (part III measured in off-medication state) at baseline *vs* week 104	NA
BIA 28–6156 (3 dose groups: 10 mg/day, 30 mg/day, and 60 mg/day) *vs* placebo	Glucocerebrosidase activator	NTR6960	1b	40 *GBA1*-PD (30 BIA 28–6156 *vs* 10 placebo)	Completed	Safety, tolerability, and immunogenicity	Good safety profile; increased intracellular glucosylceramide in PBMCs[95]
BIA-28-6156 (2 dose groups: 10 mg/day, 60 mg/day) *vs* placebo	Pharmocological chaperone	NCT05819359 (ACTIVATE)	2	237 *GBA1*-PD (target)	Active	Changes in MDS-UPDRS part II–III score at baseline *vs* week 78	NA
PR001 (LY3884961)	Gene therapy	NCT04127578 (PROPEL)	1/2a†	32 *GBA1*-PD (estimated)	Active	Safety, tolerability, and immunogenicity	NA

ADAS-Cog-13=Alzheimer's Disease Assessment Scale–cognitive subscale, version 13. CGIC=Clinician's Global Impression of Change. *GBA1*-PD=*GBA1*-associated Parkinson's disease. MDS-UPDRS=Movement Disorder Society-Unified Parkinson's Disease Rating Scale. MOCA=Montreal Cognitive Assessment. NA=not available. PBMC=peripheral blood mononuclear cells.PD=Parkinson disease. PDD=Parkinson's disease dementia. *Published results indicated by references. †Open-label clinical trial

***Table 3:* Clinical trials investigating *GBA1*-targeted treatments in people with Parkinson's disease**

people with dementia with Lewy bodies (NCT04405596), and in a phase 3 multi-centre RCT in individuals with Parkinson's disease with and without *GBA1* variants (NCT05778617).

BIA 28–6156 (previously known as LTI-291) is an allosteric activator of glucocerebrosidase that can bind glucocerebrosidase outside of the enzymatic active site, thus preventing enzymatic inhibition and promoting post-translational folding of the mutant protein. As a result, the cleavage of substrates (glucosylceramide and glucosylsphingosine) should increase, lowering the lysosomal concentrations of these substrates. No experimental studies have been published regarding the use of this compound. BIA 28–6156 entered a phase 1b trial conducted on people with *GBA1*-Parkinson's disease; the treatment over 28 consecutive days showed a positive safety, pharmacokinetic, and pharmacodynamic profile. Unexpectedly, a treatment-related transient increase in intracellular glucosylceramide in peripheral blood mononuclear cells was identified at 14 days, followed by a return to predose concentrations at 28 days; these two phases were interpreted as a result of an increased availability of cytosolic ceramide initially, as a consequence of the glucocerebrosidase activation by BIA 28–6156, followed by decreased de novo synthesis.[95] Despite the absence of clear target engagement of this compound, it has entered a phase 2 RCT in people with *GBA1*-Parkinson's disease (NCT05819359).

Non-inhibitory chaperones differ from allosteric activators, as they can improve the translocation of misfolded enzyme to the lysosome to restore enzymatic activity. A therapeutic pipeline for the identification and optimisation of chaperones of glucocerebrosidase outlines a pathway for drug development.[97]

Gene therapy and other emerging therapies

Gene therapy offers a promising strategy to restore the physiological expression of *GBA1* and its enzymatic activity in people with *GBA1*-Parkinson's disease, with satisfactory experimental evidence.[1] The first human study that evaluates intracisternal gene therapy administration in individuals with moderate to severe *GBA1*-Parkinson's disease is still at preliminary stage (NCT04127578). Results of this phase 1/2a trial are expected in 2029.

Search strategy and selection criteria

We searched PubMed on Jan 1, 2026, for articles published in English between Jan 1, 2021, and Jan 1, 2026, on *GBA1* and Parkinson disease using the terms "Parkinson's disease" AND "glucocerebrosidase" OR "GBA" OR "*GBA1*" in the title or abstract. Abstracts were scrutinised and relevant articles were selected based on content and impact.

Intravenous recombinant glucocerebrosidase, widely used in the treatment of Gaucher disease, has an excellent safety profile but no blood–brain barrier penetration owing to its large molecular weight. A phase I trial evaluating unilateral putaminal delivery of glucocerebrosidase using MR-guided focused ultrasound in four patients with *GBA1*-Parkinson's disease showed that this approach is both safe and feasible, overcoming the challenge of delivering this large protein into the brain.[98]

Further *GBA1*-targeted therapies are under evaluation in experimental models. An overview of the most promising targets is presented in the panel.[67,69–76]

Conclusions and future directions

The recognition that *GBA1* variants are the most common genetic risk factors for Parkinson's disease has provided invaluable insights into pathogenesis. Clinically, *GBA1*-Parkinson's disease is usually associated with slightly earlier onset and more rapid disease progression than idiopathic disease, particularly regarding cognitive and autonomic functions. However, it is becoming increasingly clear that clinical features are substantially influenced by the type of *GBA1* variant, and that variant type also determines the risk for the development of Parkinson's disease in asymptomatic carriers.

There are several important questions to answer in the relationship between *GBA1* variants and Parkinson's disease. Only a minority of people who carry one or even two *GBA1* variants ever develop Parkinson's disease or dementia with Lewy bodies. The understanding of the mechanisms that underlie conversion to Parkinson's disease or dementia with Lewy bodies is needed, as it is also necessary to identify the prodromal features for those carriers most at risk. The molecular and cellular effects of *GBA1* variants on the glucocerebrosidase pathway, sphingolipid metabolism, lysosomal and mitochondrial function, and their relationship to α-synuclein accumulation, aggregation, and toxicity need also to be clarified.

GBA1 appears to offer a tractable therapeutic target to modify the pathogenesis of *GBA1*-Parkinson's disease. Several trials involving several treatment strategies are underway, and more studies are planned.

Contributors

EM conducted the literature search, conceived and drafted the manuscript, and created figure 2. MT created figure 1 and critically revised the manuscript. MD, DADM, FB, DM, and ES critically revised the manuscript. AHVS conceived and drafted the manuscript. All authors read and approved the final manuscript.

Declaration of interests

DK was the Founder of Vanqua Bio and Lysosomal Therapeutics (now Bial); serves on the scientific advisory boards of The Silverstein Foundation, Intellia Therapeutics, AcureX, Vanqua Bio, and NRG; and is a Venture Partner at OrbiMed and NeuroVC. AHVS has provided paid consultancy to Capsida, Neurocrine, and Axial, is the Chief Investigator of the ambroxol phase III study, and a Principal Investigator of the MOVES-PD study. EM, MT, MD, DADM, FB, and ES declare no competing interests.

Acknowledgments

This research was funded in part by Aligning Science Across Parkinson's (grant number: ASAP-000420 and ASAP-000458) through the Michael J Fox Foundation for Parkinson's Research and the EU Joint Programme—Neurodegenerative Research (GBA-PaCTS) through the MRC grant code MR/T046007/1. ES was supported in part by the Intramural Research Program of the National Institutes of Health (NIH). The contributions of the NIH author are considered Works of the US Government. The findings and conclusions presented in this paper are those of the author and do not necessarily reflect the views of the NIH or the US Department of Health and Human Services.

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